

SwiftHaul Logistics Analysis

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GitHub: https://github.com/taloc/SwiftHaul_Logistics_Analysis

The SwiftHaul Logistics analysis is data analysis of a fictitious company, where the idea is to demonstrate the usage of Google's BigQuery and Cloud Storage.

The below summary provides the high-level results from the analysis of carrier reliability, and costs. Detailed data ingestion, storage, and the setup of dataset, tables and queries can be found in the github repository.

Data governance and access

SwiftHaul relies on Google's data governance for encryption of at rest and during transit, ensuring that our data is secured from end-to-end before the data is stored, and while we operate BigQuery for internal and external transmission of data through the internet.

For this current exercise we have IAM full access control at all levels of project, dataset, tables, and views. Permissions could be customized for limited access as needed.

Carrier reliability

The three most reliable carriers by on-time delivery are **Skyliner Air**, **SwiftHaul Ground**, and **MetroCourier**. The three least reliable are **OceanLink**, **BlueLine Freight**, and **Continental Cargo**.

carrier	total_shipments	on_time_shipmen...	on_time_rate_pct
Skyliner Air	250025	238006	95.19
SwiftHaul Ground	250269	233183	93.17
MetroCourier	249964	230329	92.14
Continental Cargo	249989	222805	89.13
BlueLine Freight	249669	217822	87.24
OceanLink	250084	203021	81.18

Lanes cost analysis

The 3 most expensive lanes by average freight-cost are:

APAC-East -> US-Central,

US-West -> APAC-South,

EU-West -> APAC-South, with an average cost of \$403.31, \$402.96, and \$402.85 respectively.

origin	destination	lane	average_cost	shipments_per_la...
APAC-East	US-Central	APAC-East -> US-Central	403.31	26719
US-West	APAC-South	US-West -> APAC-South	402.96	26706
EU-West	APAC-South	EU-West -> APAC-South	402.85	26581
APAC-East	EU-Central	APAC-East -> EU-Central	402.41	26983
US-East	APAC-East	US-East -> APAC-East	402.2	26873
US-Central	EU-Central	US-Central -> EU-Central	402.12	26637
EU-West	US-South	EU-West -> US-South	401.95	26778
US-South	EU-Central	US-South -> EU-Central	401.87	26667
US-East	EU-Central	US-East -> EU-Central	401.83	26892

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Recommendations: How do carriers compare on cost versus reliability

Carrier choice is a trade-off between cost and on-time performance. The analysis starts with the three busiest lanes, then selects carriers on those lanes with on-time performance of 90% or higher, ranked by average cost.

Top 3 busiest lanes:

Lane	Number of shipments
EU-Central -> APAC-South	27184
EU-West -> APAC-East	27081
EU-West -> EU-Central	27034

Identification of top carrier per lane based first on average cost and filtered for only those with 90%+ on-time performance.

Lane	Carrier	Average cost	On-time rate
EU-Central -> APAC-South	SwiftHaul Ground	\$249.54	93.01%
EU-West -> APAC-East	SwiftHaul Ground	\$247.89	93.01%
EU-West -> EU-Central	SwiftHaul Ground	\$155.05	92.99%

Conclusion:

SwiftHaul Ground is the lowest-cost option on all three lanes and stays above our internal 90% on-time threshold. That result should be read with care: ground freight is not the same service as air. Among air and courier providers, the next-best options are MetroCourier and Skyliner Air.